

Anakin Dey

Curriculum Vitæ

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Research Interests

Algebraic Combinatorics, Algebraic Geometry and Commutative Algebra, Computational Algebra; Algebraic and Fine-Grained Complexity, Algebraic Methods in Combinatorics, Graph Theory, Combinatorial Optimization

Education

2020–2024 **BS in Mathematics**, *University of Illinois Urbana-Champaign*, GPA: 3.91 / 4.00, Graduation with Highest Distinction
Minor in Computer Science
Deans List: Fall 2020, Fall 2021, Spring 2023

Publications

Optimizing a Model-Agnostic Measure of Graph Counterdeceptiveness via Reattachment

Anakin Dey, Sam Ruggerio, and Melkior Ornik
2023, Submitted, [arXiv:2311.15093](https://arxiv.org/abs/2311.15093)

Classifying Primitive Solvable Permutation Groups of Rank 5 and 6

Anakin Dey, Kolton O'Neal, Duc Van Khanh Tran, Camron Upshur, and Yong Yang
2023, Submitted, [arXiv:2308.13043](https://arxiv.org/abs/2308.13043)

Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency

Anakin Dey and Melkior Ornik
IEEE International Smart Cities Conference 2022, [arXiv:2206.00597](https://arxiv.org/abs/2206.00597)

Invited Talks

2023 **Classifying Primitive Groups using Computer Algebra and Change Making**
UIUC Department of Mathematics Undergraduate Friday Seminars

2023 **Classifying Primitive Solvable Permutation Groups of Rank 5 and 6**
China Three Gorges University Algebra and Combinatorics Seminar

2022 **Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency**
IEEE International Smart Cities Conference 2022

Research Experience

- Summer 2023* **Undergraduate Research Assistant**, [*REU in Algebra, Combinatorics, and Statistics*](#),
Under Prof. Yong Yang, Texas State University
- Fall 2021 – Present* **Undergraduate Research Assistant**, [*LEADCAT Lab*](#), Under Prof. Melkior Ornik,
University of Illinois Urbana-Champaign

Student Organizations

Founder and President, ACM [*SIGma*](#)
Special Interest Group in Math and Algorithms

Admin, ACM [*SIGPwny*](#)
Special Interest Group in Cybersecurity

Teaching Experience

- Spring 2021 – Present* **Undergraduate Course Assistant**, University of Illinois Urbana-Champaign
- CS 374 - Introduction to Algorithms and Models of Computation
 - Spring 2022 - Fall 2022* Course Assistant
 - Fall 2023 – Present* Course Assistant
 - CS 173 - Discrete Structures
 - Spring 2021 - Spring 2022* Course Assistant
 - Fall 2022 – Present* Head Course Assistant
 - CS 475 - Models of Computation
 - Spring 2023* Course Assistant
 - CS 125 - Introduction to Computer Science
 - Spring 2021* Course Assistant

Professional Experience

- Summer 2022* **Software Engineering Intern**, *CME Group*, Chicago, IL
- Summer 2021* **Software Engineering Intern**, *SitelQ*, Lombard, IL

Selected Coursework

- Math 500 Abstract Algebra I
- Math 501 Abstract Algebra II
- Math 502 Commutative Algebra
- Math 511 Introduction to Algebraic Geometry
- Math 518 Differentiable Manifolds I
- Math 581 Extremal Graph Theory
- CS 473 Algorithms
- CS 475 Models of Computation
- CS 598TMC Algorithms from the Fine-Grained Perspective
- CS 598MAF Algebraic and Geometric Complexity Theory
- CS 598TH1 Advances in Theoretical CS